We introduce a new database of cognate sets and etymons for the main Romance languages (Romanian, Italian, Sp

Introduction and Related work

Protoword reconstruction, consisting of recreating the words in a proto-language from their descendants in daughte
corroborating the archaeological inventory (Heggarty, 2015), and providing the basis for 'linguistic paleontology' (Epps,

In the case of Romance languages, although the mother tongue – Latin – is attested, its presence in written texts is
Furthermore, there are still numerous clearly cognate words present in several Romance languages, whose etymon is not

Although etymologists' interest in reconstructing the protolanguages has risen over the years, they still encounter nume
$C^o t' e, Russell Gray, Robert McMahon, and Mark Pagel, and co-author stook the proto word reconstruction one step furthe$

The latest computational results on Romance protoword reconstruction, in particular, are reported on the database

Starting with these remarks, our main contributions are:

We introduce a comprehensive Romance database for protoword reconstruction by processing RoBoCoP (Dinu et al., 20
We propose a strong benchmark for automatic protoword reconstruction, by applying a set of machine learning models

The rest of the paper is organized as follows: In Section 2 we present the database that we have created and offer d

Data

A major inconvenience in Historical Linguistics in general, and in computational approaches of protoword reconstru
(Dinu et al., 2024b), and determining the borrowing direction (Dinu et al., 2024a).

The ProtoRom Database

Starting with the RoBoCoP database (Dinu et al., 2023), in order to obtain cognate sets with common etymons in

[t] All cognate tuples present in the ProtoRom dataset for the Latin etymo

RO	ES	PT	IT	FR	axa
eje	axls	ASSE	als	ixrsaxls	
axrs	ASSE	ais	axaxrs	axls	
ASSE	zlls	itxlseje	axls	ASSE	
zlls	axaaxts	axls	ASSE	ars	ix
eJe	Ax1S	ASSE	zlls		

We curated the obtained data, with the help of linguists. In the process, we discarded sets that contained irrelevant

We were able to apply manual corrections for all these errors for the smaller subset of entries in the database that h

In addition to the correct series thus retained, we integrated the database created by Reinheimer-Ripeanu (2001), a

The proposed database contains a total 39,973 full or partial cognate sets along with their etymons. For the experi

A cognate set is composed of a tuple of words in different languages with a common etymon, where the tuple can b

There are 1,245 full cognate sets in the database, the rest being partial cognate sets. To facilitate distinguishing be

ProtoRom is the largest database of cognate sets for Romance languages so far, significantly exceeding the widely u

[t]

	Language combination
5,197	It–Fr
3,439	It–Es
4,605	*–It* (all except It)
1,842	Fr
3,898	Fr–Es
2,797	*–Fr* (all except Fr)
2,270	It–Ro–Pt
5,685	Ro–Es
3,394	*–Ro* (all except Ro)
2,390	It–Es–Pt
1,782	Es
4,543	*–Es* (all except Es)
2,913	Fr–Ro–Es
2,311	Ro–Es–Pt
5,202	*–Pt* (all except Pt)

sets for all languages except x.

Methodology and Experiments

Experimental Setting

For our experimental trials, we consider two settings: In the first one, we limit our dataset to only the full cognate

Data splitting. In order to train and validate our models, we split our datasets into 80

We propose a random split methodology that achieves the following properties: A Latin etymon and all of its cogna

In order to perform these splits, we construct for each Latin etymon a 5-dimensional vector (n_i) using the previous

Also note that after splitting the ProtoRom-all5 dataset, containing only the full cognate sets, we can use it as a st

Features. The proposed approaches can be split into two main categories: models for reconstructing the orthograph

Models

We use a variety of machine learning models, including classical, neural, and transformer-based (pretrained and tra

We experiment with a variety of models, including pre-trained large language models (LLMs) and current state-of-t

CRF + reranking We used an approach that relies on conditional random fields (CRFs), based on the method prop
possible protowords) and to leverage information from all modern languages. For each word in the productions list, we

Probabilistic LSTM We conducted experiments using a combination of recurrent neural networks with different dy
ProtoRom-all5 set. Like the original authors, we only apply this model on the phonemic forms of words, since the proba

Character-level transformer The next experiments conducted in this research are based on the transformer model, p

Pre-trained LLM (Flan-T5) We finally evaluate the capabilities of pretrained Large Language Models (LLMs) to so
(from 2 to 5 cognates, in the case of our experiments). For evaluation, we attempt to generate multiple output sequence

One limitation of pretrained LLMs that we cannot overcome through fine-tuning is its alphabet, which contains mo

Results

The previously described methods have been applied on both ProtoRom and ProtoRom-all5 datasets, using the ort

We also provide a comprehensive human evaluation of the results. Linguists from our team manually analyzed the

The metrics used include accuracy, (normalized) edit distance, and Cou_i, with $i \in 1, 5, 10$, which stands for an exte

ProtoRom-all5 Results

Results obtained on the ProtoRom-all5 set are shown in Table **??**. In terms of accuracy (or Cou_1), the best results

The experiments using the phonemic forms produce weaker results, with the best accuracy reaching 55.8

The probabilistic RNN models achieve very poor performances, reaching a mean edit distance of 3.11 when trained

	Accuracy			Edit/NEdit			(lr)3-5(lr)6-8 Type	
Model	Cou_1	$\text{Cou}*5$	$\text{Cou}*10$	Cou_1	$\text{Cou}*5$	$\text{Cou}*10$	Gr	Flan